MY FIRST MODEL ON HUGGING FACE

I built a model that reads bad restaurant reviews and tells you why.

“We waited 45 minutes and the burger was cold.”

wait_time 0.99 food_quality 0.83

1 day · free tools · open source

A learning project

THE PROBLEM

A 1-star review tells you that you failed. Not why.

193,166

bad reviews in one public Yelp dataset. No manager reads all of that.

Was it the wait? A rude server? Cold food? A wrong order? The bill?

WHAT IT DOES

Give it a bad review. Get every complaint inside it.

> *“Good ambiance, but the food smelled and it wasn’t worth the price for a buffet.”*
>
> `food_quality 0.89` `price_value 0.98`

8 complaint types:

`wait_time` `staff_attitude` `food_quality` `order_accuracy`
`price_value` `hygiene_safety` `ambience` `billing_policy`

Praise like “good ambiance” is not tagged. It learned the difference.

HOW I BUILT IT

Five steps, one day, $0.

01 **Write the rules**
8 labels plus rules: praise isn’t a complaint; slow isn’t rude.

02 **Label with a free LLM**
Qwen3-4B tagged 3,500 Yelp reviews. My hand check: ~18/20 right.

03 **Beat a baseline**
Simple keyword model first, as the score to beat.

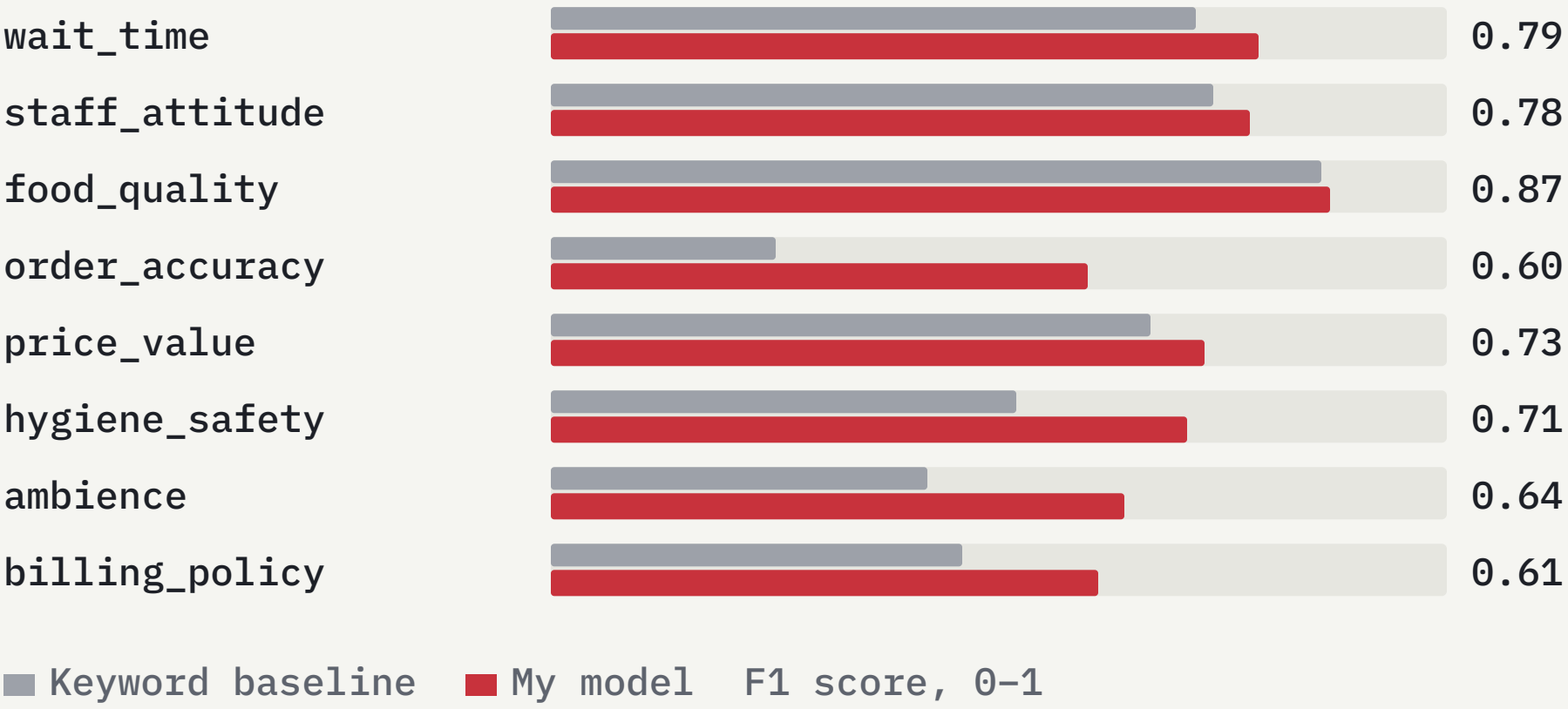
04 **Fine-tune DistilBERT**
66M parameters, weighted for rare labels, ~15 min on a free GPU.

05 **Publish honestly**
Model card with results and limitations.

RESULTS

Beats the keyword baseline on all 8 labels.

Average F1: 0.58 → 0.71. Biggest gains on rare complaints like wrong orders (0.25 → 0.60).



Test set: 354 held-out reviews labelled by the same LLM.

TOOLS AND TIME

All free. About 6 hours.

Google Colab

Free T4 GPU for labelling and training

Hugging Face Hub

Private dataset + public model hosting

Qwen3-4B-Instruct

Free open LLM that labelled 3,500 reviews

DistilBERT

66M-parameter model I fine-tuned

Python libraries

transformers, datasets, scikit-learn, pandas

Claude

AI coach for each step; git for versions

Label guide + setup	45 min
Data + pilot labelling	1.5 h
Full labelling run	40 min
Baseline + training	1 h
Checks + model card	1 h
Publish	15 min

Your own business model: plan 1–2 weeks. Most of it is collecting reviews and hand-checking labels.

The model copies its teacher.

→ Labels need written rules, or even humans disagree.

→ Rare complaints need extra weight, or the model ignores them.

→ AI-made labels carry the AI's habits. Check them by hand.

Next:

→ A bigger human-checked test set

→ Add happy and neutral reviews

→ More rare examples

→ Show the evidence

→ A weekly dashboard

TRY IT

Free on Hugging Face.

huggingface.co/nrr26/yelp-complaint-tagger

Where it could help: restaurant managers, franchises, delivery platforms, reputation agencies, restaurant software, food-safety teams.

Learning project · non-commercial license · feedback welcome